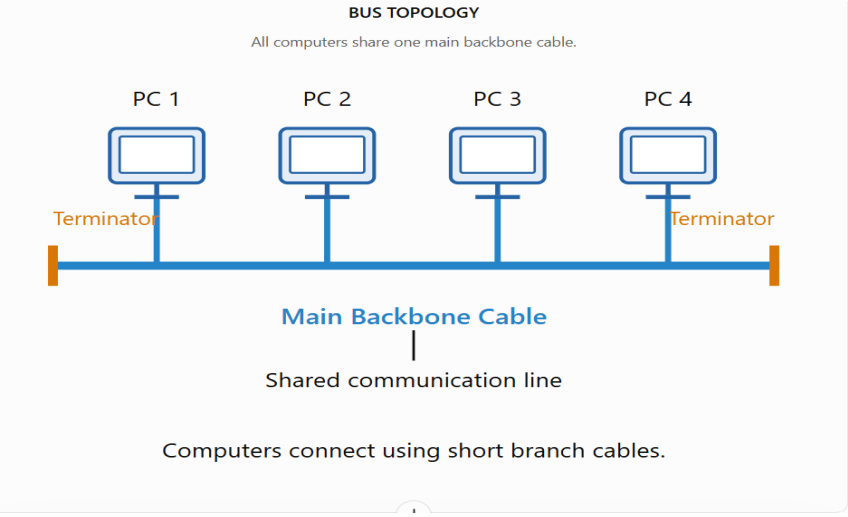
Network TopologiesTheory

1.Draw a diagram of a bus topology and label all key components, then write one real-world example where this topology might be used.

ANS:



Bus Topology Diagram

BUS TOPOLOGY

All computers share one main backbone cable.

Key Components

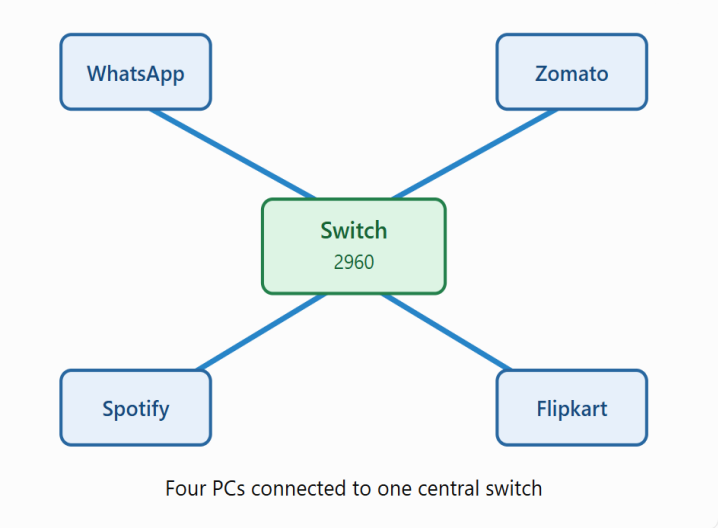
1. **Backbone cable: The main cable that carries data across the network.**
2. **Computers (PCs): Devices connected to the backbone cable.**
3. **Branch cables: Connect each computer to the main cable.**
4. **Terminators: Placed at both ends to prevent signal reflections.**

Real-World Example

A small computer lab using older Ethernet technology, such as 10BASE2, could use bus topology to connect several computers through a shared coaxial cable.

2.In Cisco Packet Tracer, create a simple star topology with one switch and four PCs connected to it; name each PC after a popular app (e.g., WhatsApp, Zomato, Spotify, Flipkart) and take a screenshot of your topology.

ANS:



Four PCs connected to one central switch

You can create this star topology in Cisco Packet Tracer by following these steps.

Star Topology — Cisco Packet Tracer

Network Diagram

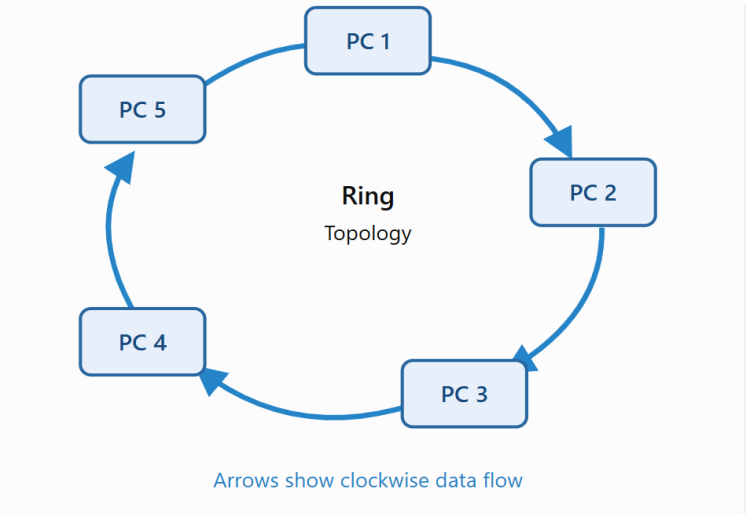
Steps in Packet Tracer

1. Open Cisco Packet Tracer.
2. Add one 2960 switch and four PCs to the workspace.
3. Rename the PCs to WhatsApp, Zomato, Spotify, and Flipkart.
4. Select Connections (lightning icon) and choose Copper Straight-Through.
5. Connect each PC's FastEthernet0 port to a separate switch port, such as Fa0/1 to Fa0/4.
6. Wait until the link lights turn green, then take a screenshot of the completed topology.

**3.Draw a ring topology diagram and explain in 2-3 sentences how data travels between devices in this setup.

Hint: Use arrows to show the direction of data flow.**

ANS:



Ring Topology Diagram

RING TOPOLOGY — CLOCKWISE DATA FLOW

How Data Travels

1. In a ring topology, each device is connected to two neighboring devices, forming a closed loop.
2. Data travels from one device to the next in the direction shown by the arrows until it reaches the destination.
3. Each device forwards the data along the ring; in some ring networks, a token controls which device can transmit.

4.Compare mesh and star topologies by listing 2 advantages and 2 disadvantages of each in a table format.

ANS:

Comparison of Mesh and Star Topologies

Topology	**Advantages (2)**	**Disadvantages (2)**
Mesh Topology	1. Highly reliable because multiple paths are available for data. 2. Failure of one link usually does not stop the entire network.	1. Expensive because it requires many cables and connections. 2. Complex to install and maintain.
Star Topology	1. Easy to install, manage, and troubleshoot. 2. Failure of one computer or cable does not affect other devices.	1. If the central switch fails, the whole network may stop working. 2. Requires more cable than some simpler topologies, such as bus.